

Executive Summary

Predicting YouTube Video Popularity using Machine Learning

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1. Business Problem

According to recent surveys, becoming a YouTuber is the #1 dream job among younger generations, surpassing traditional aspirations like doctor or engineer. With millions of videos uploaded daily, creators face intense competition to stand out and go viral.

YouTubers and digital marketers often face the challenge of predicting how their videos will perform *before* they are published. Metrics such as view count and popularity level are known only post-upload, leaving little room for optimization beforehand.

This project aims to bridge that gap by providing a data-driven predictive tool that estimates both the view count and popularity label (High, Medium, Low) based solely on metadata inputs — such as video title, tags, category, and publish time.

2. Data Understanding

I gathered video metadata from over 15 popular YouTube channels using youtube's free API, resulting in a combined dataset of ~36,000 entries. I explored key variables like title, description, tags, duration and performance metrics like viewCount, likeCount, and commentCount.

3. Data Preparation

I cleaned and transformed the data by:

- Removed duplicate entries and irrelevant columns.
- Converted data types appropriately (e.g., duration to seconds, counts to numeric).
- Handled missing values using median imputation or appropriate defaults.
- Applied **one-hot encoding** to the channelId column to convert it from a categorical identifier into a numerical format suitable for machine learning algorithms. This

approach allows the model to capture the **influence of individual channels** on video performance without imposing any false ordinal relationship between them.

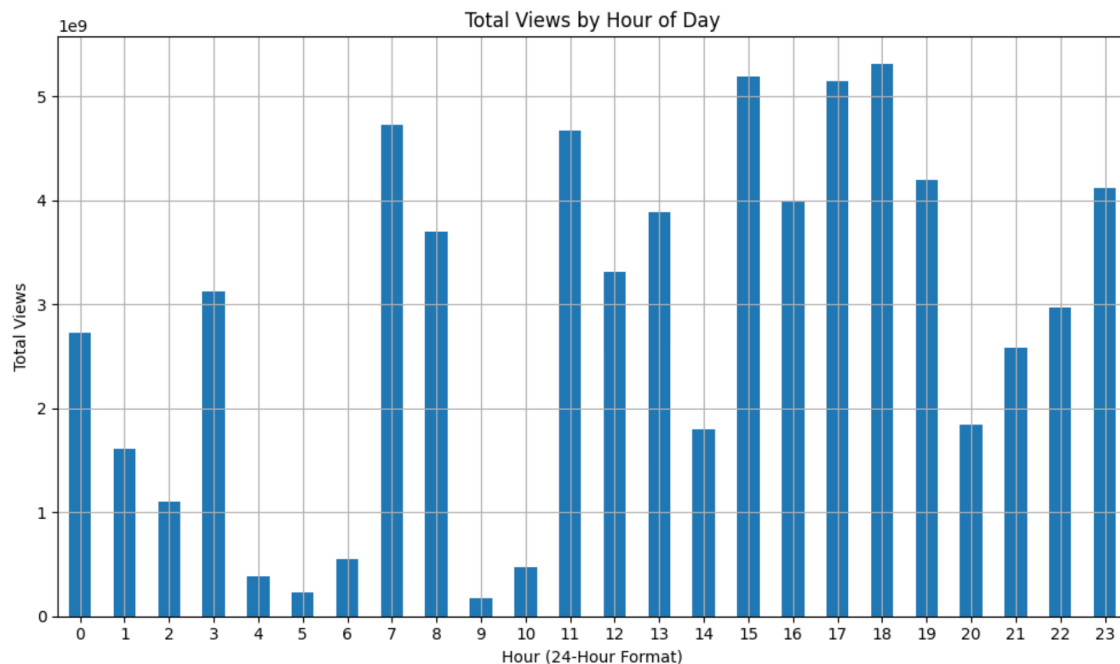
Engineered new features such as:

- Number of tags
- Length of the title
- Day of the week and hour of publishing
- Sentiment scores of titles and descriptions

4. Exploratory Data Analysis

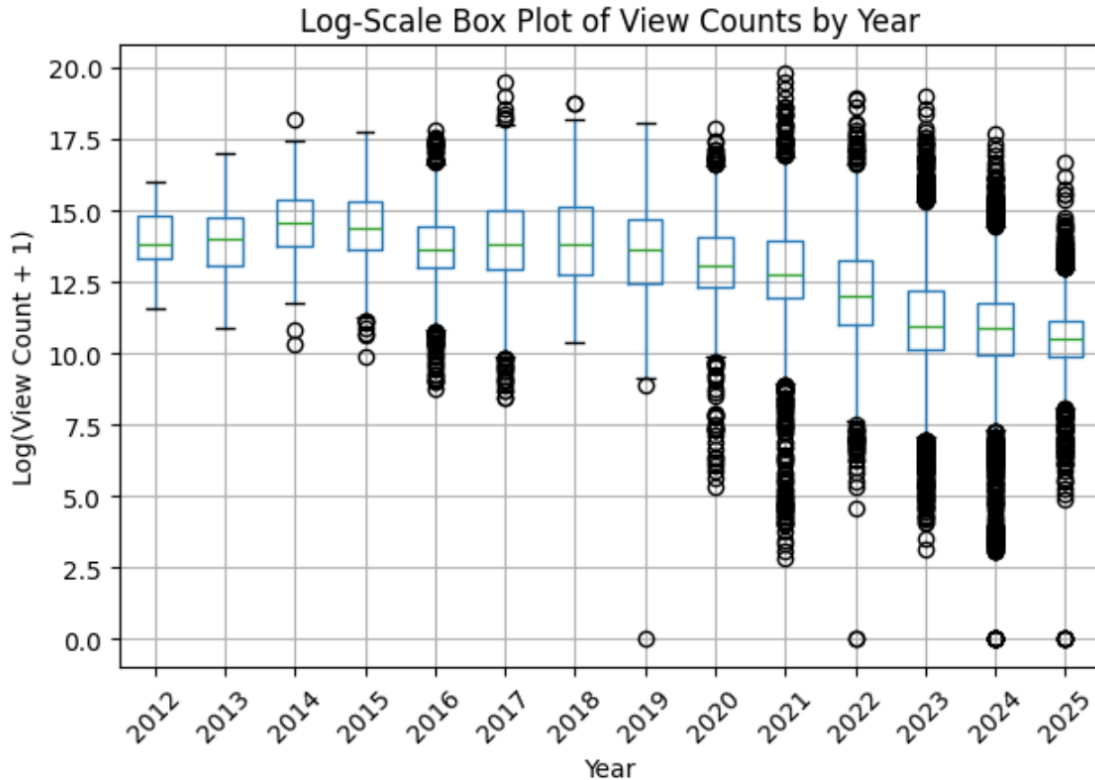
This section presents key insights derived from the dataset after performing initial data cleaning, preprocessing, and feature engineering. The analysis helped uncover important relationships between variables, which in turn informed the selection of relevant features for the regression model and guided the identification of appropriate target variables.

Views vs Hour of Upload



A bar chart of total views by upload hour showed that videos posted between 15:00 and 19:00 received the highest views, indicating peak audience activity. In contrast, uploads between 4:00 and 6:00 had the lowest engagement. These insights highlight the impact of upload timing on video performance.

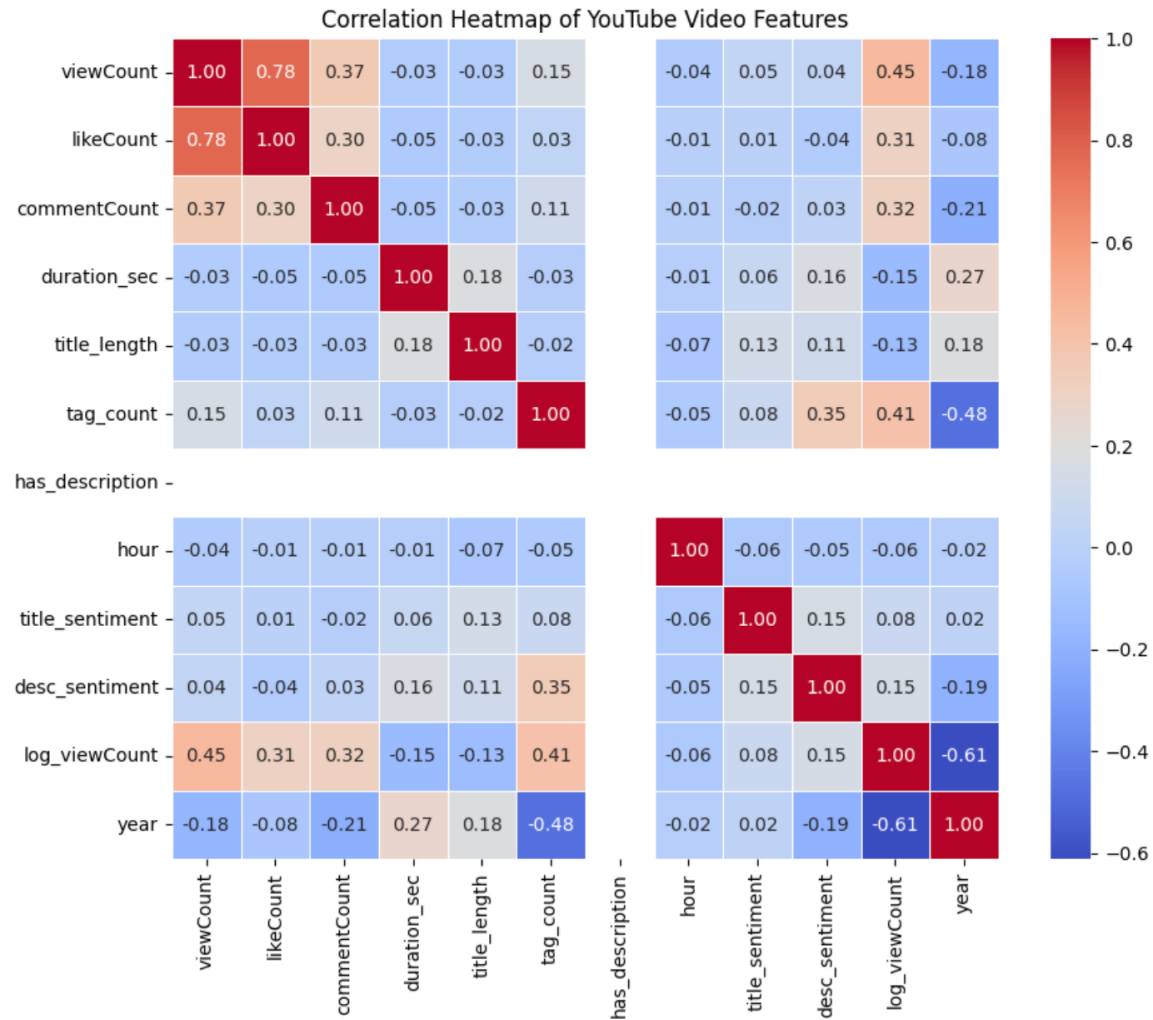
View Count Trends Over the Years



A box plot of log-transformed view counts by year showed a decline in median views for videos uploaded after 2020, possibly due to platform saturation or less time on the platform. Earlier years (2012–2018) had higher engagement, though outliers in all years suggest viral content can still emerge anytime.

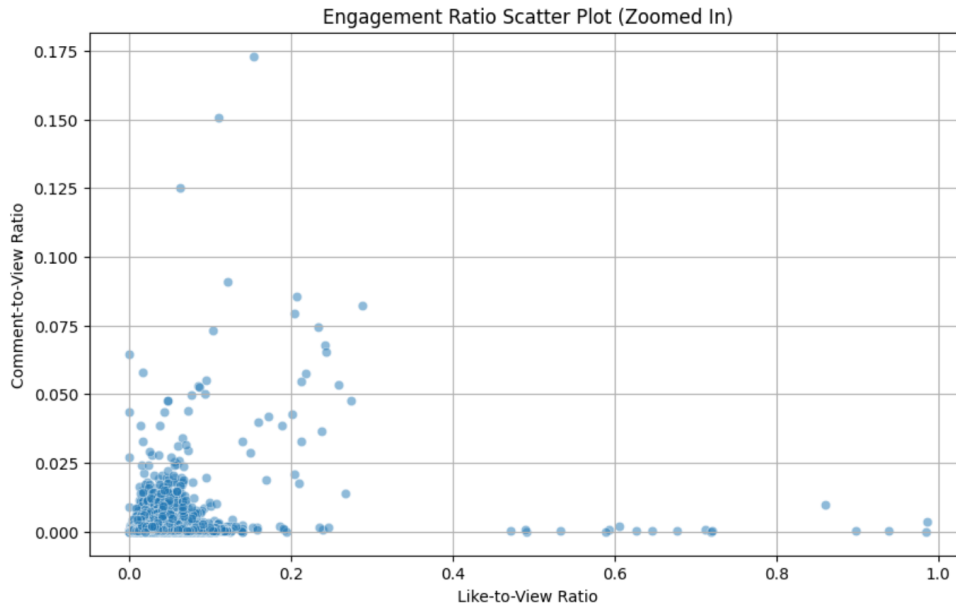
Correlation Analysis of Video Features

The correlation heatmap showed that `viewCount` is strongly correlated with `likeCount` (0.78) and `log_viewCount` (0.45), confirming likes as a key popularity signal. Moderate correlations with `tag_count` and `desc_sentiment` suggest metadata and sentiment also impact visibility. A negative correlation with `year` (−0.61) supports the decline in views for recent uploads.



Engagement Ratio Analysis

The scatter plot of like-to-view vs comment-to-view ratios (excluding outliers) showed most videos have low but consistent engagement, with like ratios below 0.2 and comment ratios below 0.05. A few outliers with high comment ratios and lower like ratios suggest conversational or controversial content. This distinction helps identify active engagement and informs content strategy or future classification.



5. Modeling Approach

Regression: Predicting video view counts

To identify the best-performing regression model for predicting video view counts, I compared *Random Forest*, *Gradient Boosting*, and *XGBoost* using *GridSearchCV*. This approach allowed me to simultaneously:

- Tune important hyperparameters
- Perform *k-fold cross-validation* (e.g., 5-fold)
- Evaluate models based on consistent performance metrics like *MSE*

Steps I followed:

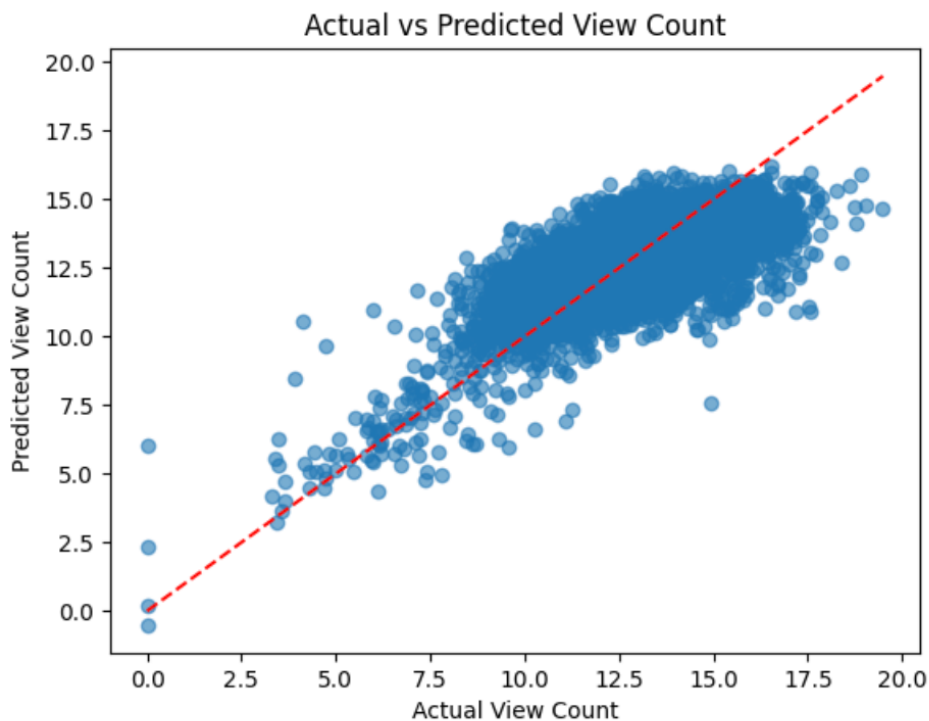
1. Defined parameter grids for each model (e.g., number of estimators, max depth, learning rate).
2. Used *GridSearchCV* to run cross-validation for each parameter combination.
3. For each model, *GridSearchCV* returned the best set of hyperparameters based on cross-validated performance.
4. Compared the mean cross-validation scores across all models to identify the best one.

Why Was Gradient Boosting Chosen?

- Gradient Boosting showed the least MSE score.
- It also handled missing values and feature interactions more effectively than the others.

This combination of **cross-validation** + **hyperparameter tuning** ensured that the final model was not only accurate but also generalizable to unseen data.

Predictive Validity



The scatter plot of actual vs predicted log view counts shows strong alignment, with most points clustering near the diagonal—indicating accurate predictions. Some variance appears at high-view outliers, but overall, the model effectively captures key patterns, validating its use for estimating video popularity.

Clustering : Assign Popularity Levels

To assign popularity levels (Low, Medium, High, Viral) to videos based on predicted view counts, I used the *KMeans clustering algorithm*. This unsupervised approach grouped videos into natural clusters without relying on manually defined thresholds.

Steps I followed:

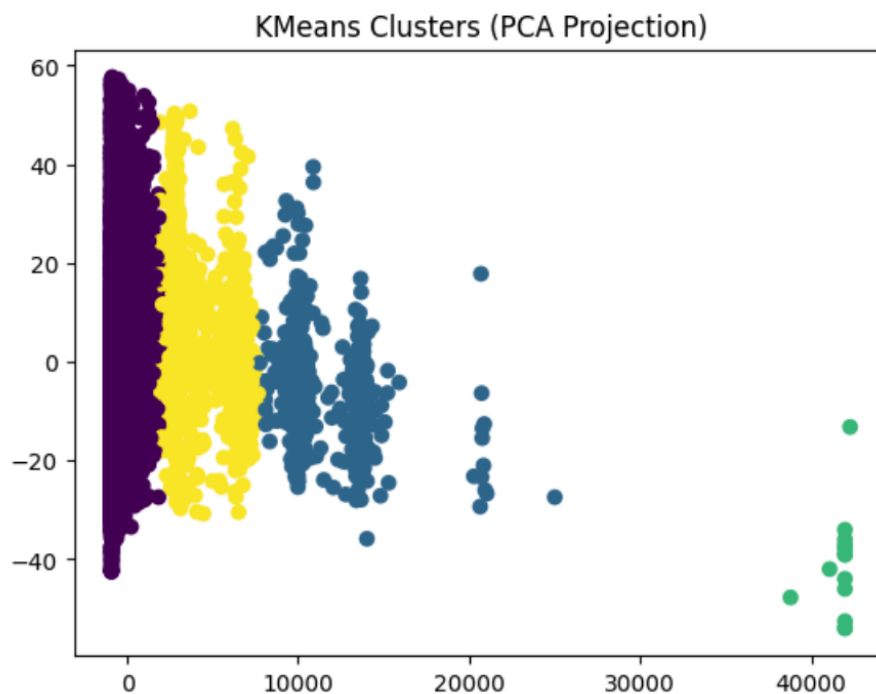
1. Applied **KMeans**, **Agglomerative Clustering**, and **GMM**
2. Evaluated each using **silhouette scores**
3. Compared cluster quality and separation visually
4. **KMeans with $k = 4$** performed best and was selected

Why was KMeans chosen ?

- It had the highest Silhouette score
- It is efficient and scalable for large datasets.
- It works well when clusters are spherical and separable in feature space.
- It's easy to interpret and apply to downstream classification tasks.

This combination of **quantitative score** and **visual confirmation** helped ensure that the final clustering structure was both meaningful and consistent.

Predictive Validity



To categorize video popularity, KMeans clustering was applied to predicted view counts. PCA visualization showed well-separated clusters, likely reflecting low to viral popularity

tiers. This approach enables data-driven segmentation without manual thresholds, supporting targeted strategies.

6. Deployment

To make the model accessible, I developed a Gradio-based interface where users can:

- Input metadata like title, description, tag count, duration and publish time.
- Get real-time predictions for: 1) Expected view count, 2) Popularity category (Low/Medium/High)

The screenshot shows a web application titled "YouTube Popularity Predictor". Below the title is a subtitle: "Enter your video's details to predict its view count and popularity category (Low to Viral)". The interface is divided into two main sections. The left section contains input fields for: "Video Title" (with the example "5 Hacks to Save Your Day!"), "Tag Count" (with the example "10"), "Description" (with the example "These are some amazing life hacks that will surprise you."), "Duration (ISO format, e.g. PT5M30S)" (with the example "PT5M30S"), "Hour of Upload" (with the example "12"), and "Channel Name" (with the example "5-Minute Crafts"). Each input field has a small icon to its right. Below these fields are two buttons: "Clear" and "Submit". The right section, labeled "output", displays the results: "Predicted View Count: 3,055,902" and "Popularity Bin: High". Below the output is a button labeled "Flag". At the bottom of the interface, there is a footer with the text "Use via API" followed by a small icon, "Built with Gradio" followed by a small icon, and "Settings" followed by a gear icon.

7. Conclusion and future enhancements

Our system offers a practical, machine learning-powered solution to a real-world problem faced by content creators. By combining regression, clustering, and sentiment analysis, and deploying through an intuitive interface, we've built a full-stack tool that enables data-informed decision-making for YouTube publishing strategies.

Future enhancements could include:

- Incorporating video thumbnail analysis.
- Tracking engagement over time.
- Expanding to other platforms like Instagram Reels or TikTok.